

NORTHERN UNIVERSITY OF BUSINESS & TECHNOLOGY KHULNA

Department of Computer Science and Engineering
Project Documentation

Course Title: Mobile Computing LAB

Course Code: CSE 4204

Project Name: Medical Report Analyzer

Document: Frontend Progress Report

Submitted To:

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Team: 02

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Submission Date: 06 July, 2026

Medical Report Analyzer

An AI-powered web application that translates complex medical reports into plain-language summaries. This document covers the current progress of the **frontend** module, built as a React single-page application that consumes the Node.js/Express backend API.

1. Team Information

Student ID	Name	Role	Task Distribution
11220320959	Hassan A.K Azad Tushin	Team Leader	AI Integration
11220320899	Farjana Easmin Minka	AI Integration Lead & Database Manager	Documentation, Database Management, Testing
11220320982	Juena Tabassum	Frontend Developer	Frontend Development: React.js
11220321050	Faika Newaz Silmi	Backend Developer	Backend Development: Node.js, Express.js

2. Frontend Technology Stack

Category	Technology	Purpose
Framework / Library	React 18.2	Component-based UI library
Build Tool	Vite 5.2	Fast dev server & production bundler
Styling	Tailwind CSS 3.4 + PostCSS + Autoprefixer	Utility-first styling, cross-browser CSS
Icons	lucide-react 0.383	Consistent icon set across UI
Data Visualization	Recharts 2.12	Trend charts (e.g. glucose trend line)
Module System	ES Modules (type: module)	Modern JS import/export syntax

3. Completed Pages & Components

3.1 Pages (via Sidebar navigation)

Page	Status	Notes
Login / Register	■ Completed	Shared AuthShell layout, form validation via Field component

Dashboard	■ Completed	Stat cards, glucose trend chart (Recharts), key markers table
Upload Report	■ Completed (UI)	Upload form wired to reportsAPI.create
Report History	■ Completed	List view with StatusBadge (Normal/High/Low/Abnormal/Monitoring)
AI Assistant (Chat)	■ Completed (UI)	Chat interface wired to chatAPI.send / history
Profile	■ Completed	Wired to profileAPI.get / update

3.2 Shared / Reusable Components

- AuthShell — centered card layout wrapper for login/register screens
- Field — labeled input with optional hint text, focus ring styling
- Logo — brand mark (light/dark variants) used in sidebar and auth pages
- Sidebar — main navigation (Dashboard, Upload, History, Assistant, Profile) with active-state highlighting and sign-out
- TopBar — page header with title, subtitle, action slot, and user avatar
- StatusBadge — color-coded pill for report/marker status (Normal / High / Low / Abnormal / Monitoring)
- Toggle — reusable animated switch component

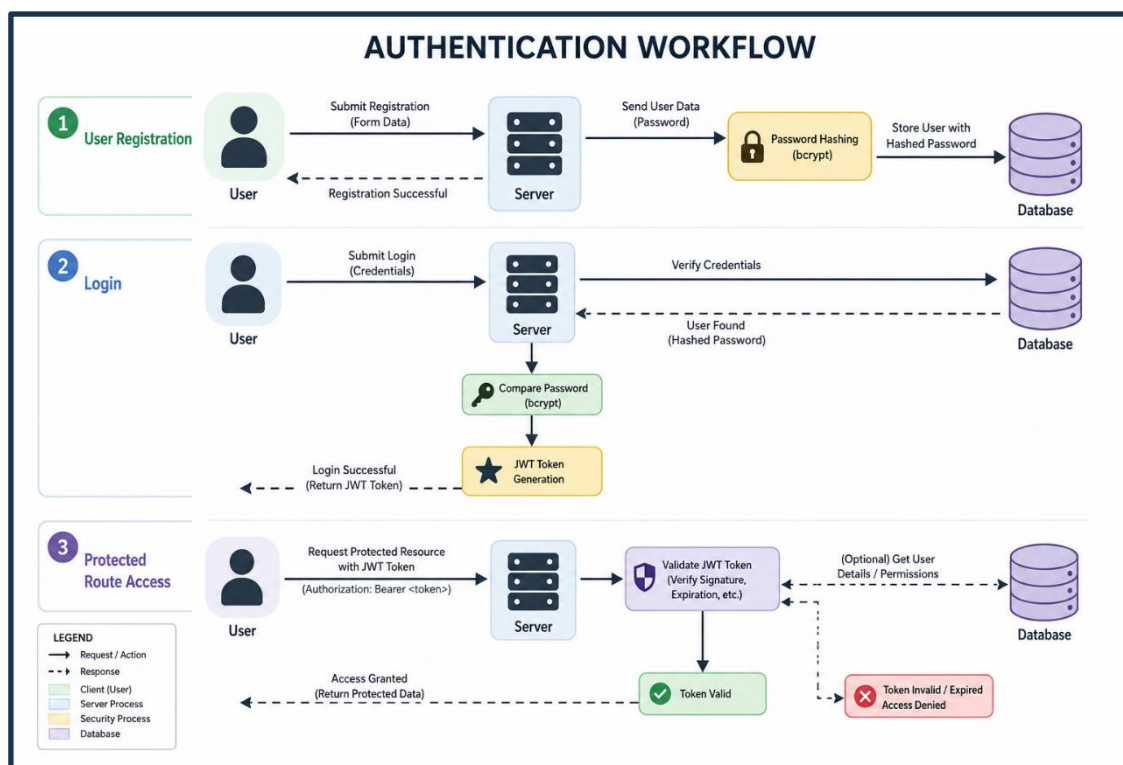
4. Backend APIs Integrated

All API communication is centralized in a single **api.js** client module, which attaches the JWT bearer token automatically, retries once on a 401 response using the refresh token, and exposes typed helper objects per resource:

Module	Endpoint	Method	Purpose
authAPI	/api/auth/register	POST	Create new account
authAPI	/api/auth/login	POST	Authenticate, store access + refresh tokens
authAPI	/api/auth/refresh	POST	Silently renew access token on expiry
authAPI	/api/auth/logout	POST	Invalidate refresh token, clear local session
profileAPI	/api/profile	GET	Fetch current user profile
profileAPI	/api/profile	PUT	Update profile fields

reportsAPI	/api/reports	POST	Upload / submit a new report for analysis
reportsAPI	/api/reports	GET	List all reports for the user
reportsAPI	/api/reports/:id	GET	Fetch a single report
reportsAPI	/api/reports/:id	DELETE	Delete a report
summariesAPI	/api/summaries/:reportId	GET	Fetch AI-generated summary for a report
chatAPI	/api/chat/:reportId	POST	Send a follow-up question to the AI assistant
chatAPI	/api/chat/:reportId/history	GET	Fetch chat history for a report

5. Authentication Flow



The frontend implements a token-based authentication flow that mirrors the backend's JWT access + refresh design:

- 1 Register / Login** — Form data (via AuthShell + Field components) is submitted to `authAPI.register / authAPI.login`.
- 2 Token storage** — On success, `access_token` and `refresh_token` are stored via `setTokens()` (`localStorage`), and the user object is cached via `setUser()`.
- 3 Authenticated requests** — Every request through the central `request()` helper attaches an

Authorization: Bearer <access_token> header automatically.

- 4 **Silent refresh** — If a request returns 401 and a refresh_token exists, the client calls /api/auth/refresh, swaps in the new access_token, and retries the original request once.
- 5 **Session expiry** — If refresh also fails, clearTokens() wipes all stored credentials and the app reloads, returning the user to the login screen.
- 6 **Logout** — authAPI.logout() invalidates the refresh token server-side and clears local storage.

6. Current Development Progress

#	Task	Status	Details
1	Project Setup	■ Completed	Vite + React scaffold, Tailwind/PostCSS configured
2	Design System / Components	■ Completed	AuthShell, Field, Logo, Sidebar, TopBar, StatusBadge, Toggle
3	Routing / Navigation	■ Completed	Sidebar-driven view switching: Dashboard, Upload, History, Assistant, Profile
4	API Client Layer	■ Completed	Centralized api.js with token attach, refresh-on-401, per-resource helpers
5	Authentication UI	■ Completed	Login/Register wired to authAPI, session persistence via localStorage
6	Dashboard & Charts	■ Completed	Recharts trend line, stat cards, marker table with status badges
7	Report Upload & History	■ Completed	Wired to reportsAPI (create/list/get/delete)
8	AI Assistant Chat UI	■ Completed	Wired to chatAPI (send/history)
9	Mock → Live Data Swap	■ In Progress	Replacing mockData.js placeholders with live backend responses
10	Responsive / Polish Pass	■ In Progress	Mobile breakpoints, loading/error states

7. GitHub Repository Link

Main Repository : <https://github.com/Tushin-vai/CSE-4204-8D-T02>

Frontend Branch: https://github.com/Tushin-vai/CSE-4204-8D-T02/tree/Frontend_Juena-Tabassum/frontend

Backend Branch: https://github.com/Tushin-vai/CSE-4204-8D-T02/tree/Backend_faikanewazsilmi/backend